- **Mass inertia** requires component-specific motion profiles in X, Y, Z and Delta
- **Advanced vision** enables multi-image inspection of large BGA ball fields
- **Coplanarity** requires precise height measurement with solder-paste thicknesses of only
100–120 µm
- **Component orientation** requires reliable Pin-1 identification

Q. How does ASMPT address challenges such as placement force, component handling, coplanarity, component/PCB warpage and Z-axis accuracy while maintaining production throughput?
- Placement Head **TWIN VHF with component-specific nozzles** and placement forces up to **100 N**
- **Component-specific mass inertia and software-controlled force profiles**
- **Heavy-Duty Conveyor** for PCBs up to **10 kg**
- **Smart Pin Support** and **PCB warpage compensation**
- **High-resolution vision and Pin-1 verification** for BGA inspection and correct orientation
- **On-Board PCB Inspection** for foreign-object detection before placement
- **3D coplanarity measurement with laser triangulation** and height resolution down to **0.5 µm**
- **In-machine spacer inspection** for presence and position with almost no impact on placement performance

Q. What are the symptoms or challenges being faced by customers which SIPLACE SX promises to overcome?
- Larger, heavier and more **complex AI-server boards**
- **Well over 20,000 placement positions**
- **Large BGAs combined with thousands of miniaturized passive components**
- **Processor BGAs up to 150 × 150 mm and >500 g**
- **10,000 to 20,000 BGA connections expected for future AI generations**
- **High-value processors make process reliability business-critical**

Q. With the increasing adoption of large, complex PCBs, what are the biggest technical challenges in maintaining consistent solder-paste deposition and print accuracy across a large printing area, and how does the DEK TQ XL address these challenges?
- **High print accuracy** and process stability across large board surfaces
- **PCBs up to 850 × 610 × 8 mm and 12 kg**, with warpage up to 4 mm
- **±20 µm @ 2.0 cpk wet-print accuracy** at a 12-second core cycle
- Automatic Smart Pin Placement with position and height verification
- Automated paste management and understencil cleaning for process stability
- Process Lens SPI + WORKS Optimization for **closed-loop offset correction**
- **Automated setup changeover** reduces the time required for manual changeovers on the printer from about four minutes to just 30 seconds

Q. As printing and placement processes generate increasingly large amounts of process data, how is ASMPT using this data to move from simply monitoring defects to predicting and preventing them?
- **Machine-integrated sensors, vision and quality functions** detect and correct deviations during production
- **Process and quality data from printing, placement and inspection** provide an end-to-end view of the SMT process
- **Process Lens SPI + WORKS Optimization** create a closed loop with automatic printing parameter corrections before deviations affect board quality
- **From inspection to prevention** through machine-integrated quality control and closed-loop process optimization

Q. Can you shed some light on WORKS Software Suite? How does it compare with an ERP or an MES suite typically being used by electronics manufacturers?
- WORKS supports **workflows around the SMT line**
- E.g. **planning, material flow, operator support, monitoring and process optimization**
- **Connecting machines, materials, operators and process data**
- **Specialized SMT functionality complementing ERP and MES**
- **Integration via standard interfaces**

Q. How does WORKS compare to other software solutions offered by ASMPT—Factory Solutions, SMT Analytics, etc?
- **Factory Solutions operates at a higher, factory-wide level** and work together with WORKS.
- **Factory Equipment Center:** factory-wide **asset and maintenance management**
- **Factory Material Manager:** holistic view and management of **material logistics**
- **SMT Analytics:** deep analysis of SMT production data, identifying patterns, deviations and optimization potential
- **Complementary solutions** rather than competing systems: WORKS > workflows around the SMT line. Factory Solutions > broader factory-level perspective and analytics

Q. What are the pain points that WORKS promises to solve for your customers?
- **Complex production planning and frequent product changes**
- **Material shortages and inefficient material flow**
- **High manual workload and limited operator support**
- **Lack of real-time transparency** into production performance
- **Process deviations and quality issues** that are difficult and time-consuming to analyze
- WORKS helps manufacturers **plan, run, monitor and continuously optimize their SMT production more efficiently**

Q. Is it correct to assume that Works will work only with ASMPT machines?
No. **Openness is a key principle of WORKS and of ASMPT**. **Open standards such as IPC-CFX and IPC-HERMES** support integration into heterogeneous production environments.

Q. AI is the talk of the hour. Are any of ASMPT solutions demonstrating the power of AI? If yes, please share details.
Yes, we are increasingly **combining AI with data analytics** to turn production and machine **data into actionable information.**
- **SMT Analytics:** identifying bottlenecks, line imbalances and optimization potential with AI-supported reporting
- **WORKS Optimization:** using AI to identify process deviations and root causes and recommend placement adjustments
- **Virtual Assist:** AI chat based on technical documentation and image-based spare-parts recognition
AI supports **data analytics, process optimization** and **practical operator support.**

Q. We are seeing a deluge of new brands (mainly from Asia) claiming high-tech features often accompanied by lower prices selling in India. How do you advice Indian customers who are confused between price, performance and brand?
As mentioned earlier, the right investment decision goes far beyond the initial investment. Customers should look at the bigger picture—overall equipment efficiency, total cost of ownership, floor space performance, and long-term investment protection. Ultimately, the smartest investment is not simply the equipment that costs less today, but the solution that continues to create greater value, flexibility, and competitiveness for years to come.

Q. Electronics manufacturers seem to be facing challenges in getting trained manpower. While you have introduced such high-tech solutions, any solutions to address their skill-set gap?
We address the skills gap through a **combination of modern training, digital assistance and easy access to support**.
- **ASMPT Academy** offers a broad training portfolio from instructor-led hands-on training and eLearning to Virtual Reality and blended learning.
- **Virtual Assist** brings expert knowledge directly into daily operations, supporting onboarding, service and maintenance. Its AI chat answers questions in natural language
- **ASMPT SMT Customer Portal** provides a central access point for additional services including ASMPT Academy training, support requests, spare parts and information about the installed machine base.

Q. Lastly, any myth amongst Indian customers that you'd like to dispel so that they make the "right choice" when selecting equipment for electronics manufacturing?
Indian customers should look beyond initial investment cost and focus on the right balance of price, performance, productivity, and long-term value—turning every investment into a stronger foundation for future competitiveness.

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